Ahmad Hossein Yazdani

ahmadyazdani@vt.edu
ayazdani1997.github.io/
Github Linkedin

Summary

PhD candidate specializing in ML systems infrastructure, HPC performance modeling, and optimization for large-scale distributed AI workloads. Experienced in diagnosing and mitigating GPU, storage, I/O, and communication bottlenecks through profiling, systems-level experimentation, and agentic AI reasoning. Background includes DOE internships and leadership-scale computing environments, including ORNL and NERSC, with recent work on AI-driven performance diagnosis for scientific workflows.

Work Experience

August,2020 – August 2026 ***Research Assistant at Distributed System and Storage Lab*, Virginia Tech**.

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science*, Virginia Tech

◦ **SODA-LLM:** a storage-aware tensor placement system for distributed LLM training (PyTorch DP/TP, DeepSpeed ZeRO), reducing **end-to-end latency by 20%** and **memory usage by 5%** on LLaMA-3-8B and Qwen-2-1.5B; **under review in Middleware 2026**.

◦ **Silo-based open framework for Federated Learning (2025-):** A **Federated Learning research** project led by Dr. Feiyi Wang (ORNL), Dr. Ali Butt (Virginia Tech), and Dr. Sahil Tyagi (ORNL). Implemented gradient compression in the gRPC communication layer for a cross-cloud/HPC Federated Learning framework, reducing cross-site communication overhead and enabling hierarchical aggregation. Achieved up to 20% latency reduction for ResNet18 trained on Cifar10

◦ **Metis** The project introduces a data access pattern for Deep Learning workloads to improve cachability. With a small cache, it improves the hit ratio by up to 4.5× compared to the LRU policy : **Shade paper at FAST23**

January,2026 – August,2026 ***Research Intern, Oak Ridge National Laboratory***.

Mentors: William Godoy, Pedro Vallero

◦ Developed **IO-SENSE**, an agentic LLM pipeline that predicts application I/O pattern shifts resulting from configuration changes (e.g., toggling Redis append mode). Benchmarking demonstrates a **90%** prediction similarity compared to actual application execution; **paper under review at IEEE Cluster 2026.**

◦ Currently developing an agentic performance-reasoning pipeline for QMCPACK to identify runtime bottlenecks and predict throughput/memory variations across workload configurations, GPU vendors, and precision modes, including mixed versus double precision execution.

June,2024 – August,2024 ***Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

◦ Examined Drishti, an HPC I/O recommendation tool; Detected many false positive warnings, and derived insights to improve the accuracy of the the Drishti I/O recommendation tool.

June,2023 – August,2023 ***Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship***.

Mentors: Suren Byna, Jean Luca Bez

◦ Identified a significant I/O variability for various HPC applications like E3SM (For earth modeling) and LAMMP (For molecular simulations) when other applications interfere with it. Later continued my research for Large AI models and language models : **SODA-LLM**

June,2021 – August,2021 ***Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group***.

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

◦ **This experience resulted in my ICDCN'25 paper, where I was able to predict the I/O pattern of the next job given the features from the past submissions of the same user with an accuracy of nearly 90% for HPC jobs.**

Computer skills

Languages Python, C, C++, pthread

ML PyTorch, DeepSpeed, scikit-learn, Hugging Face Transformers, CUDA, distributed training, LLM inference, activation checkpointing/offloading, FlashAttention/Triton exposure, Ray, Agentic AI

Cloud / Infrastructure File Systems, Chameleon Cloud gRPC, Lustre, GPFS, Slurm, MapReduce, Redis, MPI, Docker

Conference & Workshop publications

[ICDCN'25] **Yazdani, Ahmad Hossein**, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt. User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking*, ICDCN '25, page 181–190, New York, NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.

[**FAST'23★**] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL https://www.usenix.org/conference/fast23/presentation/khan.

★ Top-tier venue

Education

2020–August 2026 **PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt

May 2025 **Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt

2015–2020 : **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.